

Firms Unchained: Human Capital, Modern Services, and 21st Century Development

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We document a novel way that human capital accumulation contributes to growth and development: by allowing for the expansion of modern, multi-market service firms. In response to the recent trend of premature deindustrialization (Rodrik, 2016), economists and policymakers have looked for sources of service-led growth in low- and middle-income countries. In the high-income context of the US, Hsieh and Rossi-Hansberg (2023) show that the spread of top multi-market service firms into new locations has delivered productivity growth in services. Can LMICs replicate this model of service-led growth? We pair evidence from Brazil with a model to argue that they can, but that service firms' expansion faces a key bottleneck: the scarcity of skilled workers, who serve as managers in firms' distant establishments (Akcigit et al., 2021; Hjort et al., 2022; Engbom et al., 2024). Relieving this bottleneck allows the most productive firms to replicate production techniques across space.

We first use Brazilian data to provide descriptive evidence that motivates our analysis. In the aggregate time series, employment in multi-market firms began to grow shortly after Brazil's supply of college graduates expanded, around the year 2003. The rise of multi-market firms was concentrated in the service sector. Next, we show that skill intensity, as measured by the share of college-educated workers, grew the most within service industries. We also document that multi-market firms are significantly more skill-intensive than other formal firms, even within the same narrow industry. Skill intensity is greater for firms whose establishments operate at a longer distance from their headquarters. These facts suggest that skilled workers are more important inputs for multi-market production of services than single-market production. As our last piece of descriptive evidence, we look at wages to argue that multi-market firms are more productive than their single-market counterparts. Multi-market firms pay significantly higher wages for observationally similar workers, and their expansion into a local economy is associated with greater local wage growth, which can be explained entirely by the wage premium they pay over single-market firms.

We then supplement our descriptive evidence with a natural experiment from Brazil's expansion of college access in the late 1990s. In this period, Brazil decentralized the allocation of colleges by streamlining the entry of private colleges, subject to standardized

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accreditation checks. We use the same identification strategy as Cox (2025) and construct a microregion-level instrumental variable for local growth of college attainment, based on pre-reform college capacity relative to potential demand. In both the OLS and IV regressions, local growth of college attainment led to significant growth in the local employment share of multi-market firms. Consistent with the descriptive evidence, the growing presence of multi-market firms was concentrated in service industries. The natural experiment results enrich our understanding of the results in Cox (2025), which finds that college expansion led to local income growth and structural transformation: much of this growth was driven by the local expansion of productive, multi-market service firms.

As our next step, we are in the process of building and parameterizing a model to interpret and quantify the importance of multi-market firms for delivering gains from human capital growth. The model we have in mind combines features of Hsieh and Rossi-Hansberg (2023) and Hjort et al. (2022). Firms can operate establishments in multiple markets, and decide how much to invest in scalable production technologies. Firms' returns to these scalable technologies depend on their ability to operate establishments in distant markets. Multi-market and single-market production differ in production factor intensities. In particular, skilled workers are part of the fixed cost to operate an establishment, and this fixed cost is more intensive in skilled labor when the establishment is distant from the headquarters. As a result, expanding the local supply of skilled workers allows for more firms to operate local establishments while headquartered elsewhere. In turn, these firms respond to the opportunity to operate more establishments by increasing their investment in scalable technologies.

We plan to enrich the model with a non-homothetic demand system over modern and traditional services, as in Schwartzman (2025), to disentangle supply-driven and demand-driven expansion of multi-market firms. We will quantitatively discipline the model, partly using the empirical moments described above. Once the model is fully estimated, we will use it to answer questions such as how much stronger the productivity effect of expanding the skill supply is in the presence of multi-market firms, and how much local college expansion contributes to country-level growth through upgraded technology and organizational structure.

At this stage, we invite feedback on both the empirical and theoretical components of the paper. On the empirical side, we are interested in hearing how we can best use the observational data to make the impact of local skill supply on multi-market firm expansion as credible as possible. On the theoretical side, we invite comments on how best to set up the model to highlight and quantify the interaction between multi-market firms and local skilled labor.

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